NED UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

Department of Civil Engineering

Transportation-I

CE-302

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OPEN ENDED LAB (OEL)

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Objective:

Traffic is the foremost problem in urban living. As a traffic engineering you need to analysis the condition of traffic on busy urban arterial as define by High Capacity Manual (HCM)

Procedure: Mid-block and intersection of different arterials of Karachi

Test required: HCM defined Level of service (LOS) method

Level Of Service

Level of Service (LOS) is a qualitative measure used to evaluate the operational conditions of a transportation system and the perception of those conditions by users. It is typically applied to roadways, intersections, and transit systems to assess how well a facility or service is performing.

Key Characteristics of LOS:

1. **User Perspective:** LOS is based on factors such as speed, travel time, comfort, safety, convenience, and delay.

2. **Graded Scale:** LOS is commonly represented on a scale from A to F, where:

- A represents the best operating conditions with free-flow traffic and minimal delay.

- F represents the worst conditions with severe congestion, long delays, and poor service.

Level of Service (LOS) Criteria:

1. **Roadways:**

LOS is determined by factors such as traffic flow, vehicle speed, and density.

- LOS A: Free flow; no restrictions on maneuverability.
- LOS B: Stable flow with slight restrictions on maneuverability.
- LOS C: Stable flow but with noticeable restrictions and reduced speeds.
- LOS D: High-density flow with limited freedom to maneuver.
- LOS E: Unstable flow nearing capacity; very slow speeds and delays.
- LOS F: Breakdown conditions; traffic demand exceeds capacity.

2. **Intersections:**

LOS is determined by control delay (time spent waiting at a signal or stop).

- LOS A: Minimal delay (<10 seconds).
- LOS B-E: Gradual increases in delay.

- LOS F: Very high delay (>80 seconds for signalized intersections).

3. **Transit Systems:**

LOS considers aspects such as passenger comfort, wait times, reliability, and vehicle occupancy.

- LOS A: Comfortable seating, short wait times.
- LOS F: Overcrowding, long delays, and unreliable service.

Highway Capacity Manual (HCM)

The Highway Capacity Manual (HCM) is a vital guide in transportation engineering, developed by the Transportation Research Board (TRB), to assess the capacity, performance, and level of service (LOS) of highways, intersections, urban streets, and multimodal systems. It provides standardized methods for analyzing traffic flow, delays, and congestion, aiding in facility design, traffic management, and policy-making.

The HCM covers uninterrupted and interrupted flow facilities, multimodal analysis, and work zones, evolving through editions since 1950 to include advancements like travel time reliability and multimodal integration. Widely regarded as the industry standard, it ensures consistency in improving transportation systems while addressing modern challenges.

Use of Multi-Line Highway Level of Service Approach

Multilane highways are similar to freeways in most respects, except for a few key differences:

• Vehicles may enter or leave the roadway at at-grade intersections and driveways (multilane highways do not have full access control).

• Multilane highways may or may not be divided (by a barrier or median separating opposing directions of flow), whereas freeways are always divided.

• Traffic signals may be present.

• Design standards (such as design speeds) are sometimes lower than those for freeways.

• The visual setting and development along multilane highways are usually more distracting to drivers than in the freeway case.

Multilane highways usually have four or six lanes (both directions), have posted speed limits between 40 and 60 mi/h, and can have physical medians, medians that are two-way left-turn lanes (TWLTLs), or opposing directional volumes that may not be divided by a median at all.

Methodology Of Data Collection

To analyze traffic flow on Karsaz Road, the following method was used:

1. Video Recording:

A 30-minute video was recorded at a specific section of Karsaz Road to capture real-time traffic activity.

2. Vehicle Count in Intervals:

The 30-minute video was divided into 5-minute intervals. Vehicles were counted separately for each interval, providing detailed data on traffic flow fluctuations across smaller time periods.

3. Selection of 15-Minute Data:

Data from three consecutive 5-minute intervals (totaling 15 minutes) was selected for analysis. This selection might have been based on the most relevant or representative part of the video, such as peak traffic flow or specific time slots.

4. Application of a Factor:

The 15-minute vehicle count was doubled to estimate the total vehicle count for 30 minutes, assuming consistent traffic flow throughout the entire period.

Calculation

Level Of Service Analysis of Karsaz Road

Free-Flow Speed (FFS) Calculation

The Free-Flow Speed (FFS) is calculated using the formula:

$FFS=BFF_s-f_{LW}-f_m - f_A$

Where:
FFS = estimated free-flow speed in mi/h,
BFF_s = estimated free-flow speed, in mi/h, for base conditions,
f_{LW} = adjustment for lane width in mi/h,
f_{LC} = adjustment for lateral clearance in mi/h,
f_M = adjustment for median type in mi/h, and
f_A = adjustment for the number of access points along the roadway in mi/h.

Base Free-Flow Speed (BFF_S)

BFF_s = 50 mi/h + 7 mi/h (additional) (For Major Collector)
Therefore, BFFs = 57 mi/h.

Lane Width Adjustment (f_{LW})

From Table 7-1, for a lane width of 10 ft, f_{LW} = 6.6 mi/h.

Lateral Clearance Adjustment (f_{LC})

For a lateral clearance of 3 ft, interpolation is performed as follows:
Given:

$x_1 = 4$, $y_1 = 1.7$
$x_2 = 2$, $y_2 = 2.8$
$x = 3$
Using the interpolation formula:
$y = \frac{[(y_2 - y_1)}{(x_2 - x_1)} * (x - x_1) + y_1$

$y = [(2.8 - 1.7) / (2 - 4)] * (3 - 4) + 1.7$
$y = 2.25$
Thus, $f_{LC} = 2.25$ mi/h.

Final Free-Flow Speed

FFS = 57 - 6.6 - 2.25 - 0 = 48.15 mi/h.

Peak Hour Factor (PHF) Calculation

PHF is calculated as:

$$PHF = \frac{\text{Vh}}{(V15min\ x\ 4)}$$

Where:
V_h = Hourly traffic volume
V_{15min} = Peak 15-minute volume
N = Number of lanes

Obtained Data through Survey;

V_h = 3866 veh/hr
V_{15min} = 1042 veh/hr
N = 3 lanes
$PHF = \frac{3866}{1042*4}$
PHF = 0.92

Passenger Car Equivalent Flow Rate (V_p)

V_p is calculated as:

$$V_p = \frac{Vh}{(PHF\ x\ N\ x\ fHV\ x\ fp)}$$

f_p = 1 (daily commuters)

$$f_{HV} = \frac{1}{1+Pt(Et-1)+Prv(Er-1)}$$

where,
f_{HV} = heavy-vehicle adjustment factor,
P_t = proportion of trucks and buses in the traffic stream, P_t =8% (through survey)
P_{rv} = proportion of recreational vehicles in the traffic stream, $P_{rv} = 0$
E_t = passenger car equivalent for trucks and buses, from Table 6.6, E_t=1.5(1% upgrade)
E_r = passenger car equivalent for recreational vehicles, from Table 6.5 or 6.7.
f_{HV} = 0.9615 (as calculated from the given data)

$V_p = \frac{3866 / (0.92 \ x \ 3 \ x \ 0.9615 \ x \ 1)}{(0.92* 3* 0.9615* 1)}$

V_p = 1459 pc/ln/hr

Density Calculation

Density is calculated as:

Density = $\frac{Vp}{FFS}$

Given:

Vp = 1459 pc/hr/ln

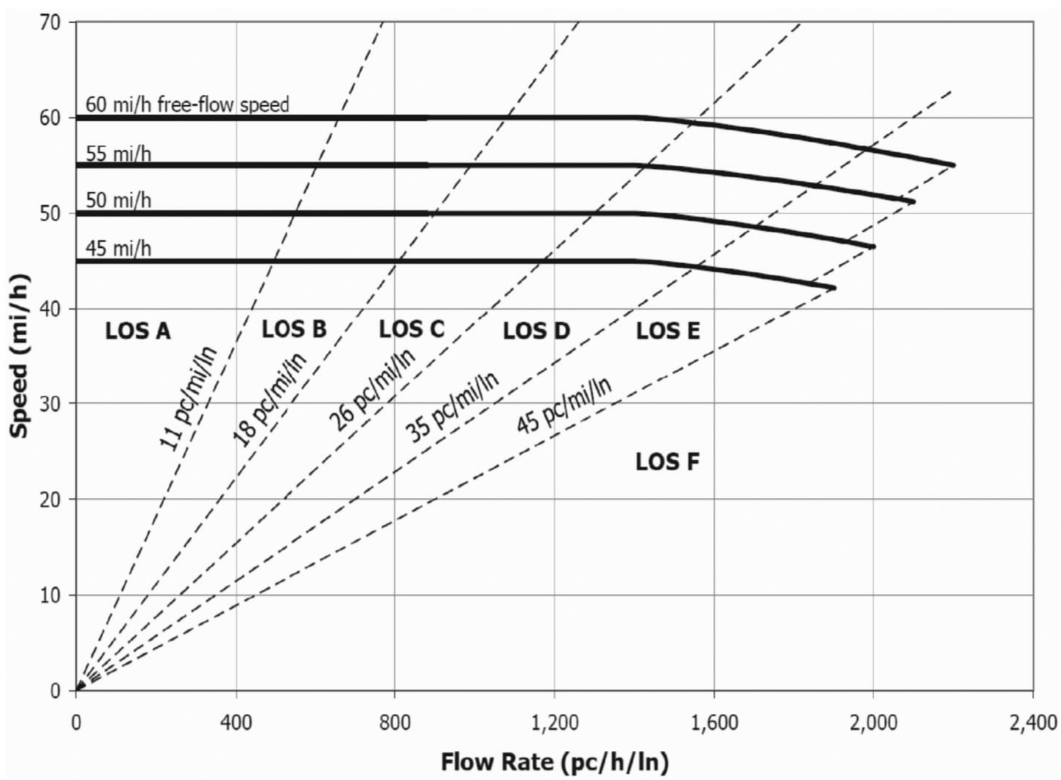
FFS = 48.15 mi/h

Density = $\frac{1459}{48.15}$

Density = 30 .3 = 35 pc/mi/ln

LEVEL OF SERVICE (LOS)

Based on the calculated Density, FFS and V_p Our Level of service comes to be **"D"**



Following Tables Were Referred:

Table 6.9 LOS Criteria for Multilane Highways

Criterion	LOS				
	A	B	C	D	E
			FFS = 60 mi/h		
Maximum density (pc/mi/ln)	11	18	26	35	40
Average speed (mi/h)	60.0	60.0	59.4	56.7	55.0
Maximum *v/c*	0.30	0.49	0.70	0.90	1.00
Maximum flow rate (pc/h/ln)	660	1080	1550	1980	2200
			FFS = 55 mi/h		
Maximum density (pc/mi/ln)	11	18	26	35	41
Average speed (mi/h)	55.0	55.0	54.9	52.9	51.2
Maximum *v/c*	0.29	0.47	0.68	0.88	1.00
Maximum flow rate (pc/h/ln)	600	990	1430	1850	2100
			FFS = 50 mi/h		
Maximum density (pc/mi/ln)	11	18	26	35	43
Average speed (mi/h)	50.0	50.0	50.0	48.9	47.5
Maximum *v/c*	0.28	0.45	0.65	0.86	1.00
Maximum flow rate (pc/h/ln)	550	900	1300	1710	2000
			FFS = 45 mi/h		
Maximum density (pc/mi/ln)	11	18	26	35	45
Average speed (mi/h)	45.0	45.0	45.0	44.4	42.2
Maximum *v/c*	0.26	0.43	0.62	0.82	1.00
Maximum flow rate (pc/h/ln)	490	810	1170	1550	1900

Note: Density is the primary determinant of LOS. Maximum flow rate values are rounded to the nearest 5 passenger cars.

Median type	Reduction in free-flow speed (mi/h)
Undivided highways	1.6
Divided highways (including TWLTLs)	0.0

Access points/ mile	Reduction in free-flow speed (mi/h)
0	0.0
10	2.5
20	5.0
30	7.5
≥ 40	10.0

Table 6.3 Adjustment for Lane Width

Lane width (ft)	Reduction in free-flow speed, f_{LW} (mi/h)
12	0.0
11	1.9
10	6.6

Table 6.6 Passenger Car Equivalents (E_T) for Trucks and Buses on Specific Upgrades

Upgrade (%)	Length (mi)	Percentage of trucks and buses								
		2	4	5	6	8	10	15	20	25
< 2	All	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
≥ 2–3	0.0–0.25	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5

Total lateral clearance* (ft)	Reduction in free-flow speed (mi/h)	
	Four-lane highways	Six-lane highways
12	0.0	0.0
10	0.4	0.4
8	0.9	0.9
6	1.3	1.3
4	1.8	1.7
2	3.6	2.8
0	5.4	3.9

Conclusion

With a vehicle flow rate of 1459 vehicles per hour and a Free-Flow Speed (FFS) of 48 mph, the Level of Service (LOS) is classified as D. This indicates that the roadway is operating at a high-density level where vehicles are still able to move but at reduced speeds compared to free-flow conditions. While traffic flow remains stable, delays and congestion are noticeable, with the facility operating near its capacity. Users may experience a decline in comfort and travel time efficiency as the traffic density increases.

Recommendations

To improve the LOS and alleviate congestion, consider measures such as optimizing signal timings, enhancing roadway capacity through lane expansions or intersection upgrades, and promoting alternative modes of transportation to reduce the demand on the facility. Additionally, traffic management strategies like congestion pricing or staggered work hours can help distribute traffic more evenly and improve overall flow.